

Is Generative AI Displacing Arabic? Perceived Change in Arabic Use Among Lebanese Arabic–English Bilinguals

Hadi Ghazi
Lebanese University
hadigghazi@gmail.com

Abstract: Generative AI tools work better in English than in Arabic, so Arabic speakers who also know English may turn to English when they use them. This study asks whether these speakers feel that AI has reduced their use of Arabic. In an anonymous online survey in Arabic, 105 Lebanese Arabic–English bilinguals who use AI regularly rated, for eight areas of daily life, whether they now use Arabic less or more because of AI. About two thirds reported using less Arabic in at least one area. The decrease was most common in work or study (half of respondents), in writing messages and in formal Arabic, and least common with family and in religion, following the order that theories of language shift predict. The more often respondents wrote to AI in English, the larger the decrease they reported, whereas how much they used AI did not matter. Simple machine-learning models identified, with moderate accuracy, who reports a decrease in formal areas, in work or study and in self-talk, but information about AI use did not make them significantly more accurate than models based on background characteristics such as age, education and English level. Respondents who felt that AI reduced their Arabic felt the same about social media, which suggests that AI continues a shift that social media began. The findings describe perceived change; tracking actual language use over time is the next step.

Index Terms: generative AI, Arabic, language shift, diglossia, perceived language change, survey, machine learning

I. INTRODUCTION

Generative AI tools perform less well in Arabic than in English. Tool-calling accuracy drops when the same tasks are prompted in Arabic [1], and four chatbots answered the same infectious-disease questions less well in Arabic than in English [2]. Bilinguals who use these tools daily may therefore do more of their AI-assisted tasks in English. Arabic was losing formal domains to English before generative AI. English dominates higher education in the UAE [3], where students at English-medium universities report weaker academic Arabic [4]; Qatar University students rate English more useful than Arabic for scientific and professional communication [5]; and in Lebanon, a small interview study found English used across education and professional life [6].

Research on generative AI and its users’ language skills has looked almost entirely at second-language learning, e.g., [7]. The closest work on Arabic records which language Tunisian users choose with ChatGPT [8] but does not ask whether that choice affects their Arabic elsewhere. To our knowledge, no study has asked Arabic speakers whether they attribute changes in their own Arabic use to AI.

This study asks Lebanese Arabic–English bilinguals in which domains, and in which direction, they report that AI has changed their use of Arabic (RQ1); whether AI-use intensity, the English share of AI use and the perceived Arabic–English quality gap predict that change (RQ2); whether the change goes with perceived difficulty using Arabic without AI and mediates the link from AI use (RQ3); and whether AI-use information improves out-of-sample prediction of who reports AI-attributed decreases beyond background characteristics (RQ4). The outcome is the change that respondents attribute to AI. It is not a measure of language loss, and it is set beside the same question asked about social media (one item).

II. BACKGROUND AND HYPOTHESES

Domains and diglossia. Language use is organised by domains such as family, friendship, religion, education and work, and a dominant language displaces a weaker one domain by domain, starting with public and formal domains [9], [10]. Arabic is diglossic: Fusha (Modern Standard Arabic) carries the formal, written functions and the spoken dialect the everyday ones [11], so formal functions should be lost first. Hence **H1**: net AI-attributed change is negative for Fusha (H1a) and for work or study (H1b); and **H2**: the decrease follows the order work/study > personal matters > family (H2a) and is larger for Fusha than for the dialect domains of self-talk, personal matters, and family and friends (H2b).

Why AI could matter. Adult first-language attrition is usually explained by reduced use of the first language and interference from an active second language [12], [13]. It has been studied mainly in migrants; in the home country the first language holds up better. Students at English-medium universities in Turkey showed no deficit on Turkish tests, although their self-rated Turkish writing fell more often than that of Turkish-medium peers [14]. Generative AI could add a new route to reduced use, following an earlier one: social networking sites moved young Arabs’ informal writing toward English and Latin script, and the resulting Arabizi has kept spreading [15]. Once access to a technology is equal, inequality moves from access to what people can do with it [16], and AI lets people do more in English. Hence **H3**: AI-use intensity (H3a), the share of AI use in English (H3b) and the perceived quality gap (H3c) predict the AI-attributed decrease; **H4**: the decrease is

associated with perceived difficulty using Arabic without AI; and **H5**: the decrease mediates the link from AI-use intensity to that difficulty.

Measuring attributed change. Wirtz and Pickl [17] showed that people can rate in an online survey how a major life event changed their language use, on a scale with a true zero (“no change”). Such retrospective attributions are only partly accurate. Perceived change after life events matched measured personality change only modestly and also tracked trends that began before the event [18]. This study adapts the format to AI and reads the results as attributions.

III. METHOD

A. Design, Procedure and Participants

An anonymous Google Form in Arabic (seven pages; 25 closed questions, five grids and one optional open question) was open from 30 September 2026, 17:57, to 4 October 2026, 21:17. Respondents were recruited through university WhatsApp groups, friends and their contacts, Instagram stories and LinkedIn posts. Before launch, a speech-language pathologist reviewed the form and approved it without changes. The Arabic wording is the instrument (Appendix). The first page described the survey as a research project at the Lebanese University, voluntary and anonymous (no names or contact details, so a response could not be withdrawn after submission) and reported only as group results; respondents confirmed being 18 or older and agreeing to take part. A screening question required Arabic as first language, use of English for study or work, and weekly AI use for six months or more.

All 105 respondents consented and passed the screening question (two began regular AI use in 2026 and may not meet the six-month criterion). Nationality was not asked: 101 grew up in Lebanon and 96 live there, all but one did one or the other, and 10 had moved country since 2022. Eighty-two were under 25 and 23 were 25–44; 60 were women; 55 held or were studying for a master’s degree or higher; 45 studied or worked in computing and 22 in engineering; 79 used AI several times a day and 37 always wrote to it in English. All 105 form the primary sample. A narrower core sample, which holds migration constant (grew up and live in the Arab region, no move since 2022; $n = 89$), is a sensitivity check.

B. Measures

The core grid asked: “Because of AI, do you now do each of these less or more?” (بسبب الذكاء الاصطناعي، هل أصبحت تقوم (بكلٍ مما يلي أقل أم أكثر؟)، comparing now with before regular AI use, from *much less* (−2) through *no change* (0) to *much more* (+2). The eight domains are shown in Fig. 1 and, as fielded, in the Appendix. A preceding grid recorded which language each activity is mostly done in today, with a “not applicable” option; a domain marked not applicable was set to missing. The *displacement score* is minus the mean change over the applicable domains (at least six), so higher means more AI-attributed decrease ($\alpha = 0.87$; McDonald’s $\omega = 0.87$, bootstrap 95% CI [0.80, 0.91] [19]; $n = 97$ complete cases). Because this consistency may partly reflect a general tendency

to blame AI, the score is read as the decrease a person attributes to AI and is not treated as a measure of domain loss.

Perceived difficulty without AI: “Because of AI, has each of these become harder or easier for you, when you use Arabic on your own without AI?” (finding the everyday word, speaking one’s dialect fluently, saying what one means fully in Arabic; higher = harder; $\alpha = 0.79$, $\omega = 0.81$ [0.67, 0.90]; 63 of 105 scored exactly 0). *AI use*: frequency, share of work or study done with AI, breadth (number of eight uses ticked), year regular use began, the language used to write to AI (English share), and exposure to AI-generated content and the language chosen for it. *AI-use intensity* is the mean of the standardised share and breadth (frequency is descriptive only: 79 of 105 are in its top category). *Perceived quality gap*: four agreement items, one reversed ($\alpha = 0.69$, $\omega = 0.70$). Also asked: a substitution item (“Has it happened that you did something in English, although you could have done it in Arabic, because AI helps you better in English?”), the same change question for social media over the same period (one item, after the AI grid), life events during the period, expected change in each domain in two years, and English level.

C. Analysis

Family A (H1–H2, within-person) used Wilcoxon signed-rank tests against zero, with exact sign tests alongside for H1a and H1b, and Page’s one-sided test for the predicted order [20], with Friedman’s test and Kendall’s W . Family B regressed the displacement score on the three AI-use measures (H3, evaluated by their joint test) and perceived difficulty on the displacement score (H4), using OLS with HC3 standard errors and t -based inference [21], adjusted for age 25 or over and starting English-medium study or work. Holm’s correction [22] was applied within each family, and a wild bootstrap (Rademacher weights, null imposed) checked the Family B p -values [23]. H5 followed PROCESS model 4 [24] with a 5000-resample percentile bootstrap; at $n = 104$ it can detect medium but not small paths [25]. A Firth logistic regression [26] of any net decrease checked H3 in binary form. Each analytic choice is reported with its alternative as a sensitivity analysis: averaging all eight domains or four of them, and adjusting for age and an English-medium start, for age only, for no covariates or for the field and AI tenure as well. Analyses were run in SPSS 23 [27] and Python [28], [29]; 34 statistics computed in both agree.

D. Machine Learning (RQ4)

The machine learning is exploratory. Targets were the number of domains with a decrease (regression), any decrease in a formal domain (work or study, or Fusha), a decrease in each domain, and three further targets (substitution, expected decrease in two years, any perceived difficulty). Features came in nested blocks: background (age 25 or over, computing field, education, English level, English-medium start), plus AI use (seven items), plus the quality gap; outcome items were never features. Models with fixed settings (L2 logistic

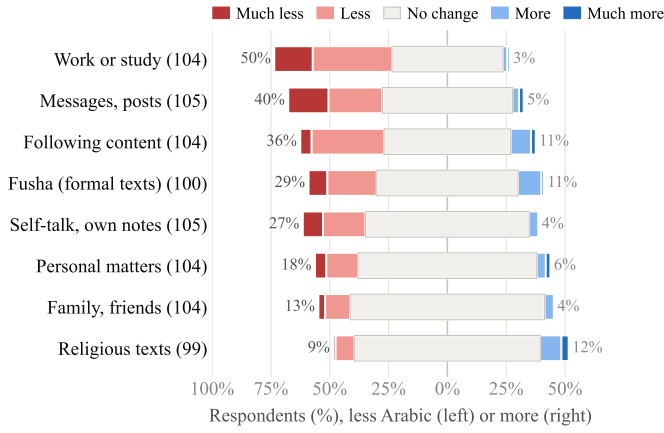


Fig. 1. AI-attributed change in Arabic use by domain (all respondents), centred on “no change”; n in parentheses excludes respondents who marked the domain not applicable.

or ridge regression as the primary model, random forest, gradient boosting) were evaluated by repeated 10×5 -fold cross-validation, with oversampling (SMOTENC [30]) inside the training folds only. Paired comparisons used the Nadeau–Bengio corrected t [31], and permutation tests (whole pipeline refit, 500 permutations) with Holm’s correction tested whether a model beat chance. Four criteria for a usable model were fixed in advance: (1) above chance after Holm’s correction; (2) over ten other random splits, mean AUC at least 0.70 and none below 0.65 (for a count, $Q^2 > 0$ throughout and a significant gain over the mean); (3) above chance without the age variable; (4) balanced accuracy at least 0.65. Models used scikit-learn and imbalanced-learn [32], [33].

IV. RESULTS

A. Where Respondents Report Change (RQ1)

Sixty-eight of the 105 respondents (65%) reported an AI-attributed decrease in at least one domain (2.2 domains on average). Decreases were most common in work or study (50% [41, 59], against 3% reporting more), writing messages and posts (40%) and following Arabic content (36%), and least common with family and friends (13%) and for religious texts, where increases (12%) outnumbered decreases (9%); for Fusha, 29 of the 100 to whom it applied reported less and 11 more (Fig. 1). More than half (59 of 105, 56% [47, 65]) reported at least sometimes doing in English what they could have done in Arabic, because AI works better in English.

B. Direction and Order (H1–H2)

The net change was negative for Fusha (H1a) and for work or study (H1b), and the decrease followed the predicted order work/study > personal matters > family (H2a; Table I). These results held in the core sample (Holm $p \leq 0.003$). Among the 25 respondents whose work or study setting did not change, decreases in work or study still outnumbered increases (8 against 1, sign test $p = 0.039$; Wilcoxon exact $p = 0.12$), and the order held ($n = 24$, Page $p = 0.011$). The decrease for Fusha was not larger than for the dialect domains (H2b).

C. Predictors and Correlates (H3–H5)

The three AI-use measures jointly predicted the displacement score, but only through the English share (Table I): respondents who wrote to AI more often in English attributed more decrease to it. The English share was Holm-significant in every specification (Table II), including with English proficiency (correlated with the share, $\rho = 0.57$) or social-media attribution controlled, and in a Firth logistic regression of any net decrease ($p = 0.004$). Intensity and the quality gap were unrelated to the score. The 23 respondents aged 25 or over reported little change (15 none; $b = -0.30$, $p = 0.002$).

H4 fell short of significance ($p = 0.053$, Holm 0.16; Spearman $\rho = 0.47$). It was positive in every specification and Holm-significant in four of the ten in Table II. One respondent outside the core sample, whose answers ran against the trend, suppressed it (Cook’s distance 0.80; without that respondent $b = 0.52$, Holm $p < 0.001$), and no other single removal brought p below the Holm threshold of 0.0125. Both measures are AI attributions collected on the same page, so H4 is not evidence that reduced use erodes ability. For H5, intensity predicted the displacement score when the English share was not controlled ($a = 0.11$, $p = 0.035$), but the path from displacement to difficulty was not significant ($p = 0.055$), the total effect was close to zero ($c = -0.03$), and the CI of the indirect effect only just excluded zero; with switching between Arabic and English as a second mediator (exploratory), it did not. Among the 82 who wrote Fusha before AI, finding formal writing harder went with more attributed decrease ($\rho = 0.46$, $p < 0.001$).

D. Social Media and Life Transitions

Respondents who reported losing Arabic to AI also reported losing it to social media ($\rho = 0.55$). Their single social-media rating showed more decrease than their AI ratings averaged over the domains (+0.48 vs. +0.27, both scored like the displacement score; $p < 0.001$) and as much as their AI ratings for the two formal domains (+0.45, $p = 0.35$), a comparison of one overall item with domain averages. The AI-attributed decrease was concentrated among the 48 who reported a social-media decrease (+0.54, against +0.04 for the other 56; exploratory), and the English share still predicted it with social-media attribution controlled (Table II). Life transitions also coincided with the AI period: 66 of the 82 respondents under 25 started university, graduated, began a new job or started studying or working in English, and the 35 who started university reported larger work/study decreases than the rest (mean change -1.03 vs. -0.42 , Mann–Whitney $p < 0.001$). Respondents whose setting did not change reported smaller ones (-0.28 vs. -0.73 , $p = 0.022$), but 11 of these 25 were aged 25 or over, and adjusted for age the difference was not significant ($p = 0.080$).

E. Prediction (RQ4)

Three linear models met all four usability criteria (Table III). Background plus AI use predicted a decrease in any formal domain (AUC 0.75; 0.76 averaged over ten other splits, range

TABLE I
CONFIRMATORY TESTS (ALL RESPONDENTS)

Hypothesis	<i>n</i>	Result	Test	Holm <i>p</i>	Verdict
<i>Family A: direction and order of AI-attributed change</i>					
H1a Fusha	100	29 less, 11 more	$z = -3.06, r = -0.31$	0.004	supported
H1b Work or study	104	52 less, 3 more	$z = -6.00, r = -0.59$	<0.001	supported
H2a Work/study > personal > family	102	means 0.64, 0.14, 0.10	Page; $\chi^2(2) = 57.06, W = 0.28$	<0.001	supported
H2b Fusha > dialect domains	100	difference +0.08	$z = 1.21$	0.22	not supported
<i>Family B: H3 predicts the displacement score; H4 predicts difficulty from it (<i>n</i> = 104, HC3)</i>					
H3 (joint)	104	three predictors	$F(3, 98) = 7.86, p < 0.001$		supported
H3a AI-use intensity		$b = -0.003 [-0.13, 0.12]$	$p = 0.96$	0.96	not supported
H3b English share of AI use		$b = 0.17 [0.09, 0.25]$	$p < 0.001$	<0.001	supported
H3c Perceived quality gap		$b = 0.06 [-0.07, 0.19]$	$p = 0.37$	0.75	not supported
H4 Difficulty without AI	104	$b = 0.37 [-0.01, 0.74]$	$p = 0.053$	0.16	not supported ^a
H5 Mediation	104	indirect 0.04 [0.0001, 0.11]	percentile bootstrap		borderline ^b

Wilcoxon tests against zero (for H1a and H1b, exact sign tests agree); H2a: Page’s one-sided test, complete-case means. Family B adjusts for age 25 or over and an English-medium start; brackets are 95% CIs. Wild bootstrap Holm *p*: H3b 0.001, H4 0.23. ^aPositive in every specification; Holm-significant in the core sample and without one influential respondent (Table II). ^bThe CI only just excludes zero; the path from displacement to difficulty is not significant ($p = 0.055$).

TABLE II
ENGLISH SHARE (H3B) AND H4 UNDER ALTERNATIVE SPECIFICATIONS

Specification	English share		H4	
	<i>b</i>	Holm <i>p</i>	<i>b</i>	Holm <i>p</i>
Primary specification	0.17	<0.001	0.37	0.160
Four-domain score	0.18	<0.001	0.34	0.028
Dialect domains only	0.17	<0.001	0.41	0.004
Age as the only covariate	0.17	<0.001	0.37	0.148
No covariates	0.16	0.001	0.38	0.110
Adding English proficiency	0.15	0.042	0.32	0.380
Adding social-media attribution	0.12	0.021	0.34	0.140
Without straight-line responders	0.15	<0.001	0.35	0.299
Without the most influential case	0.15	<0.001	0.52	<0.001
Core sample (<i>n</i> = 88)	0.17	<0.001	0.53	<0.001

Holm’s correction across the four Family B tests within each specification. Straight-line responders answered “much less” on every change row (two; one has a displacement score). The most influential case for H4 was identified by Cook’s distance after the results were seen.

0.75–0.77; balanced accuracy 0.71), in work or study (0.74; 0.74 across splits, 0.72–0.75; 0.70) and in self-talk (0.73; 0.75 across splits, 0.73–0.77; 0.65), and each stayed above chance without the age variable (permutation $p = 0.018$, 0.006 and 0.002). Writing missed the stability criterion (0.69 across splits). The number of domains with a decrease was related to the features beyond chance ($Q^2 = 0.06$, permutation $p = 0.004$) but was not predicted better than the mean ($p = 0.46$). No other target came close.

Adding AI-use information to background never improved prediction significantly (RQ4): the gain was +0.02 [−0.07, 0.10] AUC for any formal-domain decrease and +0.01 [−0.16, 0.18] Q^2 for the count (Holm $p = 1.00$ for both), and between −0.10 and +0.16 for single domains (smallest Holm p 0.27). The gain for the formal target was positive on all ten further splits (+0.03 on average), but the test could detect only gains of about 0.12 AUC. Background alone and AI use alone each reached AUC 0.70–0.73 for the formal and work/study

targets, so the two blocks overlap; English proficiency, a background feature, is correlated with the English share. The background models relied most on age, then education. For self-talk, AI use alone reached 0.75, and background alone was not above chance after correction (0.68, Holm $p = 0.072$). Random forests and gradient boosting never beat the linear model by the decision rule’s margin (0.08 AUC or 0.05 Q^2 , corrected $p < 0.05$).

Oversampling (a post hoc check) did not change these results. Six strategies applied inside the training folds (class weights, random duplication, SMOTENC, SMOTENC amplified to five times the majority class, SMOTENC with Tomek links, jittered copies) gave no reliable gain over none (best +0.02 AUC; Holm $p = 1.00$ across 24 comparisons on four targets). Applied incorrectly, to all respondents before cross-validation, they raised the apparent AUC with the amount of synthetic data, to 0.89 for any formal-domain decrease and 0.84 for substitution, whose valid AUC was at most 0.62; this measures recognition of synthetic copies, not prediction.

V. DISCUSSION

About two thirds of respondents reported that AI has reduced their Arabic in at least one domain, mainly in work and study, writing, following content and Fusha, with little in family use and none, on balance, in religion. The formal part of this profile resembles reports from English-medium education [4], [5], although here it is attributed to AI and also covers messages and media content. The order work/study > personal > family may also mirror where AI is used, and H2b, which would separate the two, was not supported: the attributions seem to follow tasks more than varieties.

AI is not the first technology to move these speakers away from Arabic. Social media shifted young Arabs’ everyday writing toward English and Latin script before generative AI [15], and respondents who reported losing Arabic to AI also reported losing it to social media. Rather than competing explanations, the two can be read as successive stages of

TABLE III
CROSS-VALIDATED PREDICTION (LINEAR MODELS, ALL RESPONDENTS)

Target (<i>n</i> yes/no; metric)	Background	+ AI use	AI use only	Δ AI use [95% CI]	Usable
Domains with a decrease (count; Q^2)	0.05	0.06	0.02	+0.01 [−0.16, 0.18]	no
Any formal-domain decrease (55/49; AUC)	0.73	0.75	0.71	+0.02 [−0.07, 0.10]	yes
Work or study (52/52; AUC)	0.73	0.74	0.70	+0.00 [−0.08, 0.08]	yes
Writing messages and posts (42/63; AUC)	0.71	0.70	0.62	−0.01 [−0.09, 0.06]	no
Self-talk and own notes (28/77; AUC)	0.68	0.73	0.75	+0.06 [−0.04, 0.15]	yes
Fusha (29/71; AUC)	0.64	0.68	0.67	+0.03 [−0.06, 0.13]	no
<i>Further targets, criteria fixed before they were modelled</i>					
Substitution (59/46; AUC)	0.56	0.61	0.62	+0.05 [−0.07, 0.18]	no
Expected decrease in two years (count; Q^2)	0.02	−0.01	−0.01	−0.03 [−0.19, 0.14]	no
Any difficulty without AI (34/71; AUC)	0.66	0.60	0.53	−0.06 [−0.16, 0.04]	no

Means over repeated 10×5-fold cross-validation; chance is AUC 0.50 and $Q^2 = 0$ (negative: worse than predicting the mean). Δ : background plus AI use minus background, Nadeau–Bengio CI. Usable: all four criteria of Section III-D, for background plus AI use. Other single domains (personal matters, family, religion, following content): background plus AI use AUC 0.50–0.69, none usable. Holm across all 13 targets modelled: any formal-domain decrease, work or study and self-talk 0.026.

one technology-driven shift: social media moved informal communication toward English, and AI carries the shift into work, study and writing, while AI-generated content may add to what people already follow online. AI also leaves its own mark. The English share of AI use predicted the decrease in every specification, including with social-media attribution controlled, whereas the amount of AI use and the perceived quality gap did not: what goes with the decrease is the language in which people use AI, not how much they use it. Because the outcome is an attribution and the AI period also covers the move to university or work, which shapes perceived language change [17], the size of AI’s own contribution is a question for longitudinal designs.

The usable models identify, within this sample, who reports AI-attributed decreases in formal domains, work or study and self-talk, at an AUC of about 0.75. Validated internally, they are a starting point for external validation in larger samples. AI use added no significant gain over background, so RQ4 is not supported, although small gains cannot be excluded. Oversampling adds no information, and done before the split it produces spurious accuracy, an error behind near-perfect results in other studies [34], [35].

Limitations and future work. The sample was recruited through university, personal and social-media networks and is young, highly educated and computing-heavy, so the percentages describe this group rather than all Lebanese bilinguals. Apart from the expert review, the instrument was not back-translated or formally pretested, and change was rated retrospectively, over two to four years for most respondents (94 of 105 began regular AI use in 2024 or earlier). Future work can build on these results with larger and more varied samples, repeated measures that track AI and social-media attributions over time to test whether the two reinforce each other, external validation of the models, and behavioural records of the language people use with AI.

VI. CONCLUSION

The Lebanese bilinguals in this study report that generative AI has reduced their Arabic in work or study, in writing and in Fusha, and those who use AI more in English report more decrease. The pattern is consistent with AI extending a shift that social media began, carrying it from everyday communication into work, study and writing. Simple models identify who reports such decreases, although AI-use information added no significant gain over background characteristics. Longitudinal designs and behavioural measures can now test how much of this shift AI itself drives.

DATA AND CODE AVAILABILITY

The Arabic and English instrument, analysis code and outputs are at <https://github.com/hadigghazi/genai-arabic-displacement>. Individual responses are not shared because participants were told that results would be reported only in aggregate.

APPENDIX: KEY ITEMS AS FIELDIED

<i>Because of AI, do you now do each of these less or more?</i>	بسبب الذكاء الاصطناعي، هل أصبحت تقوم بكل مما يلي أقل أم أكثر؟
Using Arabic in my work or studies	استخدام العربية في عملي أو دراسي
Writing messages or posts in Arabic	كتابة الرسائل أو المنشورات بالعربية
Using Arabic when I talk to myself or write my own notes	استخدام العربية عندما أحدث نفسي أو أكتب ملاحظاتي الخاصة
Using Arabic for personal matters	استخدام العربية في الحديث عن أموري الشخصية
Using Arabic with my family or friends	استخدام العربية مع عائلتي أو أصدقائي
Reading or writing Fusha (news, documents, articles)	قراءة الفصحى أو كتابتها (أخبار، وثائق، مقالات)
Reading or listening to religious texts	قراءة نصوص دينية أو الاستماع إليها
Following content in Arabic (videos, articles)	متابعة محتوى بالعربية (فيديوهات، مقالات)

REFERENCES

- [1] K. Kubrak, A. El-Moselhy, A. Alsulami, R. Altuwaim, H. I. Fawaz, and F. Alsaby, “Arabic prompts with English tools: A benchmark,” in *Proc. 2025 IEEE Int. Conf. Big Data (BigData)*, 2025, pp. 5452–5461.
- [2] M. Sallam, K. Al-Mahzoum, O. Alshuaib *et al.*, “Language discrepancies in the performance of generative artificial intelligence models: An examination of infectious disease queries in English and Arabic,” *BMC Infectious Diseases*, vol. 24, no. 1, 2024, art. no. 799.

- [3] A. Al-Bataineh, "Language policy in higher education in the United Arab Emirates: Proficiency, choices and the future of Arabic," *Language Policy*, vol. 20, no. 2, pp. 215–236, 2021.
- [4] T. I. Masri, "A critical examination of the Arab undergraduate students' perceptions of their academic Arabic proficiency in three EMI universities in the UAE," *English Language Teaching*, vol. 12, no. 4, p. 21, 2019.
- [5] E. Mustafawi, K. Shaaban, T. Khwaleh, and K. Ata, "Perceptions and attitudes of Qatar University students regarding the utility of Arabic and English in communication and education in Qatar," *Language Policy*, vol. 21, no. 1, pp. 75–119, 2022.
- [6] M. Mehio, "Upholding and resisting English language ideologies: Identity negotiation of five Sunni Muslims in Lebanon," *Journal of Language, Identity & Education*, vol. 25, no. 5, pp. 2132–2146, 2026.
- [7] M. Saarela, S. Gunasekara, and P. Kumarage, "A meta-analysis of generative AI effects on language proficiency and affective-cognitive outcomes in language learning," *Discover Computing*, vol. 29, no. 1, 2026, art. no. 116.
- [8] A. Bouzayenne, "Language choice and AI-mediated code-switching in Tunisians' interactions with ChatGPT: Age-related and task-domain variation," *Dialogica*, vol. 2, no. S1, pp. 33–54, 2026.
- [9] J. A. Fishman, "Who speaks what language to whom and when?" *La Linguistique*, vol. 1, no. 2, pp. 67–88, 1965.
- [10] J. A. Fishman, "Domains and the relationship between micro- and macrosociolinguistics," in *Directions in Sociolinguistics: The Ethnography of Communication*, J. J. Gumperz and D. Hymes, Eds. New York, NY, USA: Holt, Rinehart and Winston, 1972, pp. 435–453.
- [11] C. A. Ferguson, "Diglossia," *Word*, vol. 15, no. 2, pp. 325–340, 1959.
- [12] A. Chaouch-Orozco and F. Martín-Villena, "Network science reveals the early signs of L1 lexical attrition: Introducing the Lexical Attrition Foundation (LeAF) framework," *Bilingualism: Language and Cognition*, vol. 28, no. 1, pp. 43–53, 2025.
- [13] F. Gallo, B. Bermúdez-Margaretto, A. Malyshevskaya *et al.*, "Lexico-semantic attrition of native language: Evidence from Russian-Hebrew bilinguals," *Language Learning*, vol. 75, no. 3, pp. 771–800, 2025.
- [14] A. Ayçiçeği-Dinn, S. Şişman-Bal, and C. L. Caldwell-Harris, "Does attending an English-language university diminish abilities in the native language? Data from Turkey," *Applied Linguistics*, 2015, advance online publication.
- [15] H. Alghamdi and E. Petraki, "Arabizi in Saudi Arabia: A deviant form of language or simply a form of expression?" *Social Sciences*, vol. 7, no. 9, 2018, art. no. 155.
- [16] E. Hargittai, "Second-level digital divide: Differences in people's online skills," *First Monday*, vol. 7, no. 4, 2002.
- [17] M. A. Wirtz and S. Pickl, "Major life events as drivers of perceived linguistic change across adulthood," *Language Variation and Change*, vol. 37, no. 1, pp. 87–110, 2025.
- [18] T. Schwaba, J. J. A. Denissen, M. Luhmann, C. J. Hopwood, and W. Bleidorn, "Subjective experiences of life events match individual differences in personality development," *Journal of Personality and Social Psychology*, vol. 125, no. 5, pp. 1136–1156, 2023.
- [19] T. J. Dunn, T. Baguley, and V. Brunson, "From alpha to omega: A practical solution to the pervasive problem of internal consistency estimation," *British Journal of Psychology*, vol. 105, no. 3, pp. 399–412, 2014.
- [20] E. B. Page, "Ordered hypotheses for multiple treatments: A significance test for linear ranks," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 216–230, 1963.
- [21] J. S. Long and L. H. Ervin, "Using heteroscedasticity consistent standard errors in the linear regression model," *The American Statistician*, vol. 54, no. 3, pp. 217–224, 2000.
- [22] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [23] R. Davidson and E. Flachaire, "The wild bootstrap, tamed at last," *Journal of Econometrics*, vol. 146, no. 1, pp. 162–169, 2008.
- [24] A. F. Hayes, *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*, 3rd ed. New York, NY, USA: Guilford Press, 2022.
- [25] M. S. Fritz and D. P. MacKinnon, "Required sample size to detect the mediated effect," *Psychological Science*, vol. 18, no. 3, pp. 233–239, 2007.
- [26] D. Firth, "Bias reduction of maximum likelihood estimates," *Biometrika*, vol. 80, no. 1, pp. 27–38, 1993.
- [27] *IBM SPSS Statistics for Windows, Version 23.0*, IBM Corp., Armonk, NY, USA, 2015.
- [28] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with Python," in *Proc. 9th Python in Science Conf.*, 2010, pp. 92–96.
- [29] P. Virtanen, R. Gommers, T. E. Oliphant *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, no. 3, pp. 261–272, 2020.
- [30] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [31] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.
- [32] F. Pedregosa, G. Varoquaux, A. Gramfort *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [33] G. Lemaitre, F. Nogueira, and C. K. Aridas, "Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning," *Journal of Machine Learning Research*, vol. 18, no. 17, pp. 1–5, 2017.
- [34] G. Vandewiele, I. Dehaene, G. Kovács *et al.*, "Overly optimistic prediction results on imbalanced data: A case study of flaws and benefits when applying over-sampling," *Artificial Intelligence in Medicine*, vol. 111, 2021, art. no. 101987.
- [35] M. S. Santos, J. P. Soares, P. H. Abreu, H. Araujo, and J. Santos, "Cross-validation for imbalanced datasets: Avoiding overoptimistic and overfitting approaches," *IEEE Computational Intelligence Magazine*, vol. 13, no. 4, pp. 59–76, 2018.